Table 4. Regime-conditioned within-family selection used after Definition 4.1 chooses a watermark family. For each estimated edit rate $\hat{\varepsilon}$ and watermark candidate m, the concrete scheme is the method with the highest score $S(m, \hat{\varepsilon})$.

Family	Estimated Edit Rates	Edit rate conditioned Score $S(m, \hat{\varepsilon})$ with tuple (AUC, z)	Selected Rep.	Watermarking Parameters
Token-level	$\hat{\varepsilon} \approx 0.25$ $\hat{\varepsilon}_s \approx 0.06$	**HCW: 1.137 (0.910, 3.40)** DiPMark: 1.104 (0.900, 3.90) HeavyWater: 1.072 (0.880, 4.20) SimplexWater: 1.052 (0.870, 4.50) Unigram: 0.982 (0.880, 8.80) KGW: 0.954 (0.860, 9.60)	**HCW**	δ-reweight variant (`method=delta`)
Semantic	$\hat{\varepsilon} \approx 0.42$ $\hat{\varepsilon}_s \approx 0.10$	**PMark: 1.305 (0.850, 1.20)** SemStamp: 1.220 (0.820, 1.50) SimMark: 1.170 (0.800, 1.70)	**PMark**	Rejection parameter $\rho = 0.25$
Dist.-preserving	$\hat{\varepsilon} \approx 0$ $\hat{\varepsilon}_s \approx 0$	**CGW: 1.990 (0.990, -5.80)**	**CGW**	Security parameter $\lambda = 128$, fixed secret key across runs

Note. This table explains the second stage of the Hybrid. Definition 4.1 first selects the family from the estimated edit regime. The concrete scheme is then chosen within that family using the regime-conditioned score $S(m, \hat{\varepsilon}) = \mathrm{AUC}(m, \hat{\varepsilon}) + \frac{1}{1+\max(z(m,\hat{\varepsilon}),0)}$. Greener cells indicate higher within-family scores.